

SAVEETHA SCHOOL OF ENGINEERING*SIMATS ENGINEERING | DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING***CO5 ASSESSMENT TOOL 2 — SEMINAR ON EMERGING TECHNOLOGIES****Intelligent AI-Based Airport Baggage Tracking and Management System***An Engineering Framework Integrating RFID, IoT Sensing, Predictive AI Analytics, and Autonomous Logistics*

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Student Declaration & Code of Conduct:

I, **RR Thiyagesh** (Reg. No. **192572007**), hereby certify that this seminar report is an original compilation of my technical study. I have strictly adhered to the academic integrity guidelines specified for the CSA1017 Software Engineering course assessment.

Signature: _____**Date:** August 22, 2026

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1. INTRODUCTION TO THE EMERGING TECHNOLOGY

Modern commercial aviation hubs operate under immense operational strain, handling upwards of 4.5 billion passenger bags annually across complex international transport networks. As global flight traffic continues its post-pandemic expansion, airport logistics systems are reaching physical and technological boundaries. Traditional Baggage Handling Systems (BHS) rely heavily on optical barcode scanning, centralized legacy databases, and static conveyor routing trees—a paradigm that inherently suffers from visual line-of-sight failures, delayed data reporting, and an inability to dynamically adjust to physical bottlenecks.

The **Intelligent AI-Based Airport Baggage Tracking and Management System** represents a transformative paradigm shift in software engineering and airport operations. By converging Radio Frequency Identification (RFID), Internet of Things (IoT) edge telemetry, Cloud-native event-driven microservices, Machine Learning (ML), and Computer Vision (CV), this technology replaces passive logging with proactive real-time orchestration.

Core System Paradigm: Shifting baggage operations from reactive troubleshooting (repatriating lost bags after missed flights) to automated predictive management (forecasting bottlenecks 30 minutes in advance and rerouting bags dynamically).

From a software engineering vantage point, this architecture is designed as a fault-tolerant distributed system capable of ingesting high-throughput sensor data streams (exceeding 20,000 events per second). It applies deep learning models to predict mishandling risks while executing graph optimization algorithms to control high-speed physical conveyor diverters in real time.

2. INDUSTRY CONTEXT & NEED FOR AUTOMATION

According to annual industry statistics published by SITA, mishandled luggage—comprising delayed, lost, damaged, or misrouted bags—costs the global aviation industry more than \$2.2 billion each year in direct compensation, repatriation freight fees, manual handling labor, and customer service operations.

2.1 Critical Bottlenecks in Conventional BHS

- **Optical Barcode Limitations:** Traditional laser scanners require direct line-of-sight to reads tags. Creased labels, inverted luggage orientation, or poor ambient lighting lead to read failure rates between 10% and 15%, forcing manual intervention.
- **Black-Box Blind Spots:** Legacy systems only capture baggage data at fixed physical scan checkpoints (check-in, main sorter, loading bay). In transit between these points, the physical position of the asset is completely untracked.
- **Static Conveyor Routing:** Conventional sorters use hardcoded decision trees. If a conveyor belt jams or a chute reaches capacity, bags continue to be dispatched into the jam, exacerbating operational delays.
- **High Connection Mishandling:** Transfer connections account for over 60% of all mishandled bags globally. Tight layover windows leave zero margin for sorting delays or manual transport mistakes.

2.2 Problem Statement & Primary Objectives

Problem Statement: Baggage delays and misrouting severely disrupt passenger travel and inflate airline overheads. Existing infrastructure lacks real-time visibility and predictive analytics. An AI-driven baggage platform must track assets continuously, optimize conveyor paths, predict layover delays, alert passengers automatically, and supply actionable analytics to airport operations.

System Objective	Software / Hardware Strategy	Target Operational Metric
Real-Time Tracking	UHF RAIN RFID passive tags & Edge Tunnel Readers	99.8% tag read accuracy across terminals
Dynamic Routing	Dijkstra/A* shortest-path algorithms on conveyor graphs	35% reduction in total sorting transit time
Predictive Delays	LSTM Recurrent Neural Networks on flight & queue data	Predict layover delays 30 mins in advance
Passenger Alerts	MQTT / WebSockets push notification microservices	Instant mobile push latency (< 2 seconds)
Misroute Interception	Computer vision cross-validation at sorter chutes	> 70% reduction in misrouted bags

3. SYSTEM ARCHITECTURE & WORKING PRINCIPLE

The platform is engineered around an event-driven IoT architecture. Baggage telemetry flows sequentially through physical sensing, edge ingestion, real-time stream processing, AI prediction, and automated actuator dispatch.

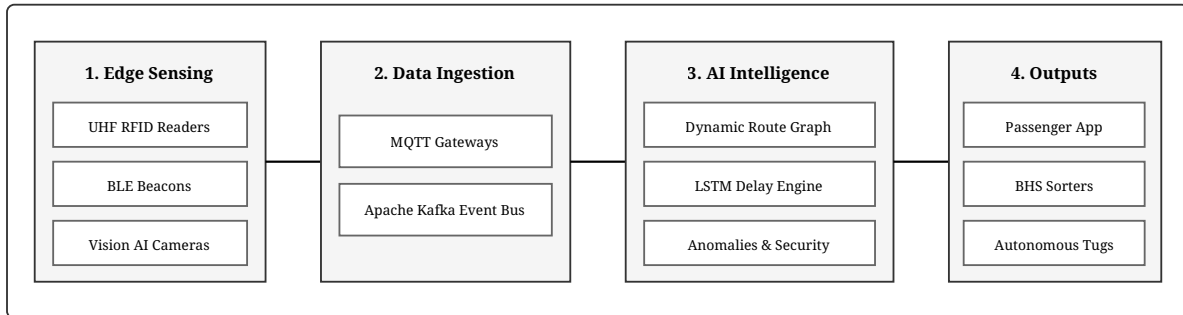


Figure 1: High-level architectural pipeline detailing data flow from physical IoT sensing to distributed microservices.

3.1 End-to-End Operational Sequence

- 1. Check-in Encoding:** A smart thermal bag tag embedded with a RAIN RFID chip is printed and encoded with the passenger's 10-digit IATA License Plate Code (LPC).
- 2. Conveyor Reading:** As luggage moves along main lines, overhead tunnel readers energize the passive tags via electromagnetic induction, scanning data without requiring line-of-sight.
- 3. Event Publication:** Raw reader events (containing `Tag_ID`, `Reader_ID`, `Timestamp`, `RSSI`) are published to a Kafka cluster.
- 4. AI Decision Execution:** The routing engine continuously computes optimal conveyor paths and dispatches instructions to pneumatic diverter arms.
- 5. Aircraft Manifest Verification:** Handheld RFID readers used by ramp staff confirm every bag matches the flight manifest prior to container loading.

4. SOFTWARE STACK & HARDWARE INFRASTRUCTURE

Implementing an enterprise-grade AI tracking platform demands tight coupling between industrial-grade edge hardware and resilient cloud microservices.

4.1 Hardware Deployment Componentry

- **RAIN RFID UHF Readers & Antennas:** Fixed industrial RFID readers (Impinj Revolution / Zebra FX9600) installed at conveyor intersections, sorting arrays, and loading docks operating within 860–960 MHz UHF bands.
- **Active BLE Gateways:** Bluetooth Low Energy beacons placed throughout arrival halls and apron yards to track ground handling carts and autonomous electric tugs.
- **Optical CV Array:** Overhead industrial cameras executing YOLOv8 models to record bag physical dimensions, inspect handles for tears, and verify optical barcodes as a fallback.
- **Autonomous Guided Vehicles (AGVs):** LiDAR-equipped self-driving luggage carts dispatched dynamically for urgent short-connect transfers across aprons.

4.2 Distributed Software Architecture

Microservice Layer	Technology Implementation	Engineering Function
Ingestion Gateway	Node.js, EMQX MQTT Broker	Handles 20,000+ persistent socket connections from RFID edge nodes.
Stream Engine	Apache Kafka, Apache Flink	Deduplicates raw tag reads and constructs windowed telemetry streams.
AI Core & Inference	Python, PyTorch, Ray, Scikit-Learn	Executes pathfinding graph optimization and LSTM delay predictions.
Storage Subsystem	PostgreSQL, TimescaleDB, Redis	Manages flight manifests, time-series sensor logs, and cache states.
API Gateway	Go (Golang), gRPC, WebSockets	Exposes low-latency REST endpoints and dispatches push notifications.

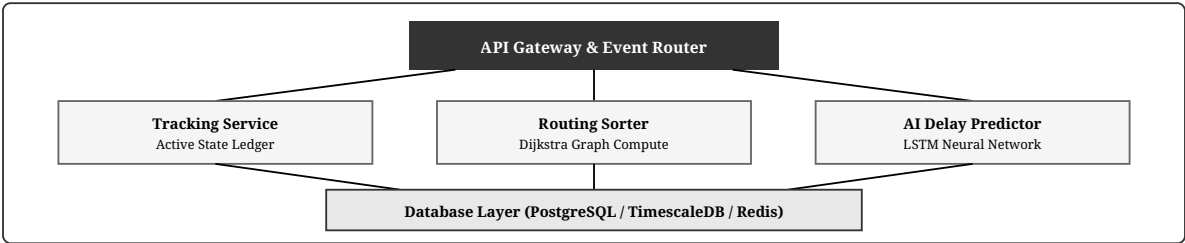


Figure 2: Microservice decoupling and database persistence layout.

5. PRACTICAL APPLICATIONS & OPERATIONAL WORKFLOWS

The integration of intelligent baggage tracking directly impacts multiple operational workflows across terminal landside, airside sorting, and flight loading zones.

5.1 Real-Time Passenger Push Notifications

Rather than experiencing post-flight anxiety at baggage carousels, passengers receive proactive tracking milestones pushed directly to airline mobile apps via WebSockets:

- 08:15 AM — Baggage enrolled at Counter 12, Terminal 2.
- 08:31 AM — Cleared TSA automated explosive detection screening.
- 08:52 AM — Loaded into ULD Container #4812 for Flight AA-204.
- 11:40 AM — Arrived at hub terminal; routed to Carousel 4 (Est. arrival: 6 mins).

5.2 Short-Connect Transfer Management

Inbound flights arriving behind schedule create high risk for transfer luggage missing connecting flights. The AI routing engine automatically flags short-connect bags upon touchdown, instructing sorters to route them to express chutes where autonomous electric tugs transport them across the apron directly to departing gates.

5.3 Security Interception & Automated Diversion

When customs or security agencies flag a bag for manual screening, its status is updated in the database within milliseconds. As the bag passes the next conveyor diverter, the physical arm immediately redirects it into a secure holding bay, automatically preventing it from being loaded onto the aircraft.

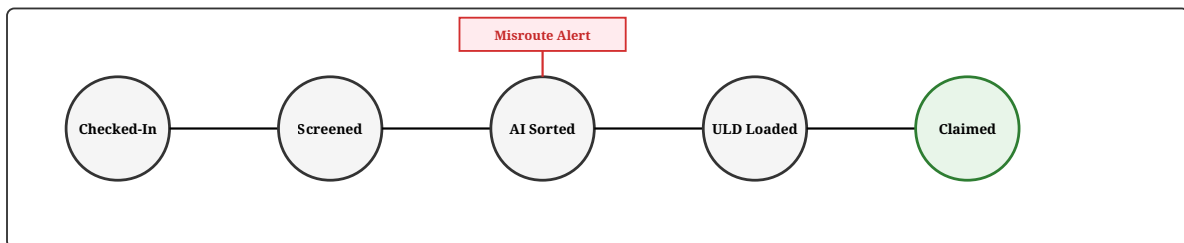


Figure 3: Baggage state machine transition model from intake to passenger reclamation.

6. PREDICTIVE ANALYTICS & ROUTE OPTIMIZATION ENGINEERING

The primary algorithmic strength of this platform relies on dynamic graph traversal combined with deep learning sequence prediction models.

6.1 Conveyor Graph Optimization Algorithm

The physical conveyor network is modeled as a directed graph $G = (V, E)$ where vertices V represent diverter junctions and edges E represent conveyor belts. The dynamic edge weight $W(e)$ is evaluated continuously:

$$W(e) = \alpha \cdot (Length_e / Speed_e) + \beta \cdot (Active_Load_e / Max_Capacity_e) + \gamma \cdot Maintenance_Penalty$$

If a conveyor belt experiences a mechanical breakdown or jam ($Active_Load \rightarrow Max_Capacity$), $W(e)$ approaches infinity. The routing service immediately reroutes subsequent bags along secondary conveyor loops using Dijkstra's shortest path algorithm.

6.2 Predictive Delay Model (LSTM Neural Network)

A Long Short-Term Memory (LSTM) network evaluates historical flight performance, weather factors, and internal conveyor queue states to calculate the probability of a bag missing its connection.

Model Parameter	Telemetry Source	Analytical Significance
Arrival Delay (ΔT_{arr})	ADS-B Flight Tracking Radar	Measures remaining runway-to-gate transfer window.
Ramp Transit Distance (D_{ramp})	GIS Apron Map / GPS Data	Estimates driving time between arrival and departure gates.
Conveyor Queue Factor (L_{bhs})	IoT Optical Sensing Array	Quantifies physical queue bottlenecks in sorting tunnels.
Weather Severity (W_{index})	METAR Meteorological API	Adjusts ramp handling times due to extreme weather conditions.

The model outputs a probability $P(Delay) \in [0, 1]$. When $P(Delay) > 0.75$, the platform automatically dispatches an express transport request to the ramp management team.

7. OPERATIONAL TRADE-OFFS & COMPARATIVE ANALYSIS

While AI-based smart baggage management significantly improves key airport metrics, system implementation requires balancing capital costs against operational efficiency gain.

7.1 Key System Benefits

- **Drastic Loss Reduction:** Non-line-of-sight RFID tracking reduces misrouted and lost baggage by up to 75%, yielding major financial savings for partner airlines.
- **Improved Customer Satisfaction:** Granular mobile app tracking provides transparency, reducing passenger anxiety and terminal queueing times.
- **Automated Regulatory Compliance:** Seamlessly fulfills IATA Resolution 753 mandates, which enforce strict custody tracking at four key operational milestones.

7.2 Engineering Challenges & Limitations

- **Capital Expenditure (CapEx):** Outfitting existing legacy facilities with thousands of RFID antennas and cloud infrastructure requires high upfront capital.
- **Electromagnetic Reflections (EMI):** Heavy metallic structures in airport basements can reflect RF signals, necessitating precise antenna tuning and filtering algorithms.
- **Cybersecurity Risks:** Aggregating real-time passenger flight data requires robust Zero Trust network architecture and AES-256 data encryption.

7.3 Architectural Comparison

Operational Metric	Legacy Barcode BHS	Smart AI / RFID Platform
Scan Success Rate	85% – 90% (Line-of-sight dependent)	99.5% – 99.9% (RAIN RFID non-contact)
Routing Strategy	Static decision trees	Dynamic graph pathfinding
Bottleneck Handling	Reactive (Manual clearing)	Predictive (Reroutes 30m prior)
Passenger Visibility	Check-in desk only	Continuous real-time app notifications
Misroute Rate	~6.5 bags per 1,000 passengers	< 1.2 bags per 1,000 passengers

8. EMERGING INDUSTRY TRENDS & FUTURE HORIZONS

The modernization of aviation logistics is accelerating through advancements in edge computing, robotics, and digital twin simulation platforms.

8.1 Recent Technological Advances

- **Mandated RFID Adoption:** Major international hub airports are transitioning from legacy barcode paper tags to standard RFID inlay tags under IATA mandates.
- **Volumetric Computer Vision Profiling:** In-motion 3D vision systems measure luggage volume and weight, flagging irregular bags before they reach delicate conveyor chutes.
- **Digital Twin Airport Modeling:** Physics-based virtual replicas (built using NVIDIA Omniverse) simulate airport baggage flows under extreme peak holiday loads to identify bottleneck zones safely.

8.2 Future Engineering Scope

Next-Generation Horizon: Integrating Swarm AI for autonomous ramp fleets, Tagless Biometric Baggage Recognition (identifying bags by microscopic surface wear signatures), and ultra-low latency 5G private networks.

Private 5G network installations across airport aprons will allow edge RFID readers and self-driving luggage tugs to communicate with sub-millisecond latency, removing the need for costly physical cables across runway ramps.

9. CASE STUDY, CONCLUSION & REFERENCES

Empirical Benchmark: Global Aviation Hub Deployment

A major international airport handling over 60 million passengers annually upgraded its central terminal with an AI-driven RFID tracking platform paired with an Apache Kafka streaming backend.

- **Observed Outcomes:**

- Baggage read accuracy improved from 88.2% to **99.7%** within 90 days.
- Transfer mishandling dropped by **68%**, generating \$14.5 million in annual cost savings.
- Passenger satisfaction scores regarding baggage delivery time increased by **42%**.

10. CONCLUSION

The **Intelligent AI-Based Airport Baggage Tracking and Management System** illustrates the practical value of applying modern software engineering principles to complex logistics infrastructure. By combining RFID edge sensing, real-time microservices, dynamic graph optimization, and predictive AI analytics, the platform eliminates structural blind spots in baggage handling. Implementing these technologies directly addresses global mishandling costs, improves airport efficiency, and delivers a transparent, stress-free experience for air travelers.

11. ACADEMIC & INDUSTRY REFERENCES

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